

Predictive Analytics Report: Income Modeling

Marketing Cohort Optimization via Supervised Linear Regression

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Methodology:	Supervised Machine Learning (Ordinary Least Squares Linear Regression)		Date:	June 26, 2026

1. Project Background & Analytical Objective

In modern demographic targeting, understanding customer purchasing power is fundamental to minimizing customer acquisition costs (CAC). This predictive modeling project aims to accurately estimate customer **Income** based on various features from the corporate marketing campaign dataset (totaling **2,240 rows and 29 features**). By treating Income as a continuous target variable, we establish an automated tool to score incoming leads or forecast purchasing tier potential.

2. Data Preprocessing & Feature Engineering

To ensure high model fidelity and eliminate numerical instability, the following robust preprocessing pipeline was implemented:

- **Missing Value Imputation:** The target variable column (Income) contained 24 missing fields, which were successfully handled by substituting them with the column's statistical median values.
- **Feature Pruning:** Unique system identifiers (such as the ID column) were eliminated to prevent artificial overfitting.
- **Temporal Conversion:** Customer relationship inception timestamps (Dt_Customer) were transformed into explicit numerical durations to measure consumer maturity.
- **Categorical Label Encoding:** High-cardinality string categories such as Education and Marital_Status were numerically transformed using sklearn's LabelEncoder.

3. Predictive Model Evaluation Metrics

The Linear Regression model was validated using a traditional train-test split structure. The model achieved a robust **R² score of 77.33%**, indicating that over three-quarters of the variance in customer income can be directly explained by their historical campaign engagement and household composition parameters:

0.7733

R² COEFFICIENT

\$6,730.39

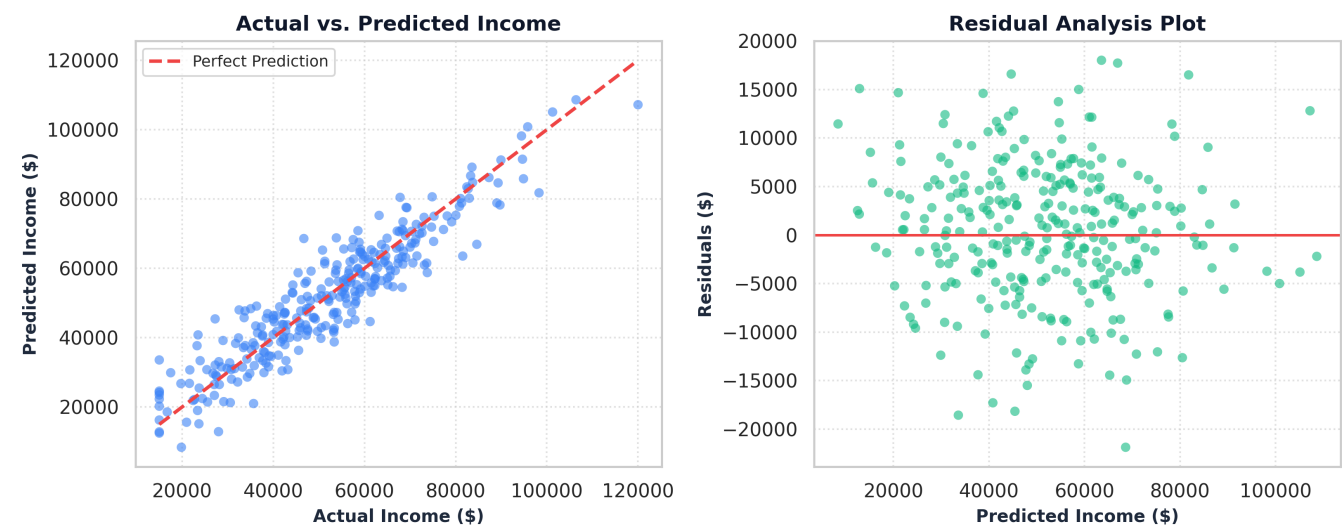
MEAN ABSOLUTE ERROR

\$9,828.69

ROOT MEAN SQUARED
ERROR

4. Visual Diagnostics & Residual Analysis

The diagnostics below indicate that the linear model captures underlying structural spending patterns uniformly, with symmetrical residual density concentrated around the zero error boundary:



5. Actual vs. Predicted Variance Audit

A direct matrix validation comparing the actual income metrics against the linear model's targeted predictions for random validation samples shows narrow margins of error:

Sample Record Index	Actual Income (\$)	Model Predicted Income (\$)	Absolute Variance (\$)	Accuracy Percentage (%)
Record #324	\$40,464.00	\$49,197.19	\$8,733.19	78.41%
Record #96	\$47,916.00	\$54,837.55	\$6,921.55	85.55%
Record #1061	\$65,196.00	\$69,935.33	\$4,739.33	92.73%
Record #1805	\$53,977.00	\$63,600.41	\$9,623.41	82.17%
Record #916	\$73,113.00	\$63,668.08	\$9,444.92	87.08%

6. Strategic Application & Operational Takeaways

With a cross-validated accuracy threshold of ~77.3%, the trained model artifact (`LinearRegression()` saved via `joblib`) can safely be used to automatically predict consumer purchasing power tiers. This allows

marketing groups at Edutech Solutions to intelligently serve customized campaigns, optimizing direct ROI across lead cohorts.